Does network depth track deduction depth?

L*(T) measured at three problem sizes · 49523504, 49524030 · 2026-10-25

No. In every series where L* can be measured, it is FLAT in T.

N=24 rho=0.75: L* = 3 · N=24 rho=0.6: L* = 3 · N=48 rho=0.75: L* = 3 · N=48 rho=0.6: L* = 4

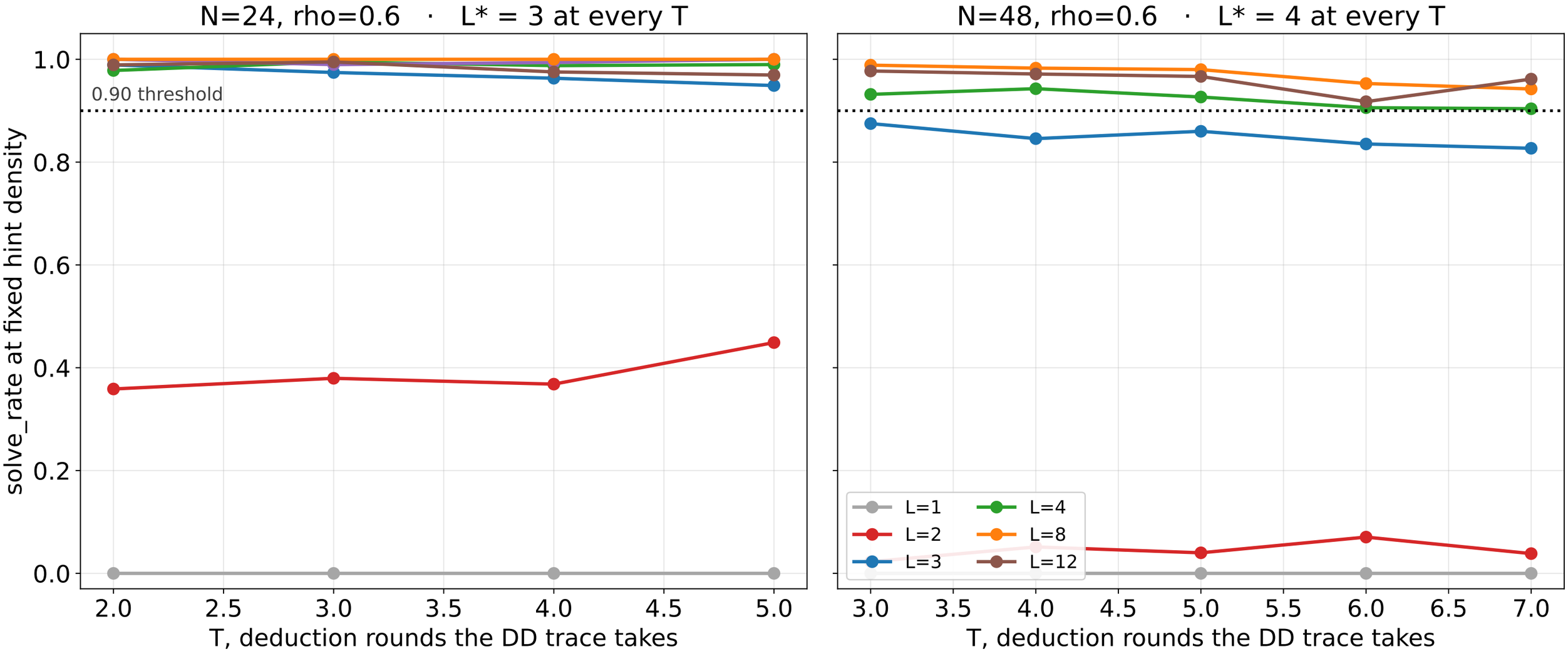
15 depth bins, 4 series, 0 of them rising. What moves L* is density
and problem size -- never the number of deduction rounds inside a series.

New this run: the num_layers=3 cell at N=48 (the one the scan skipped), which is what lets L* be compared across N at all; and an N=96 probe that found the recipe's ceiling below the 0.90 threshold, so L* is not measurable there and the 36 h scan stays unsubmitted.

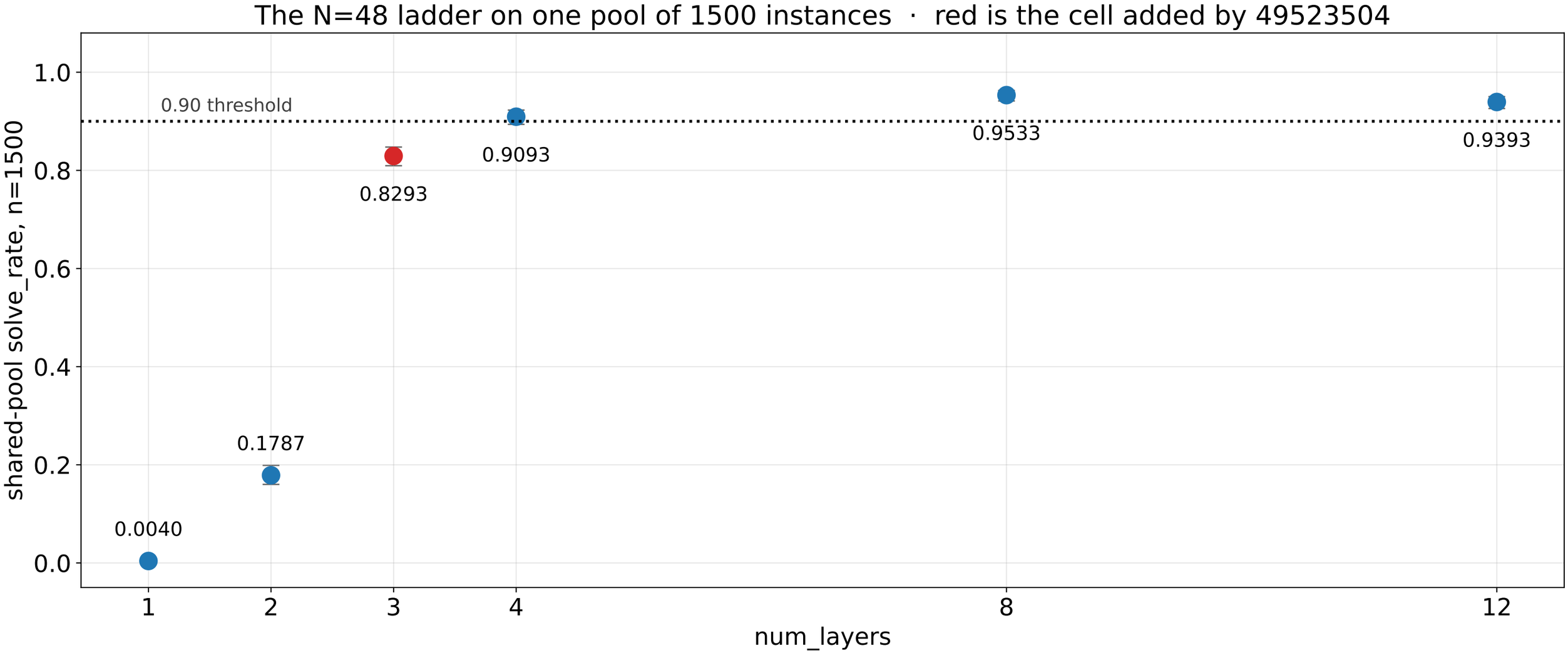
L*(T): the smallest num_layers reaching solve_rate ≥ 0.90 on depth-T instances

size	density	T bins	L grid searched	L*(T)	flat in T?
N=24	rho=0.75	1-3	{1,2,3,4,6,8,12}	3, 3, 3	**yes, flat**
N=24	rho=0.6	2-5	{1,2,3,4,6,8,12}	3, 3, 3, 3	**yes, flat**
N=48	rho=0.75	2-4	{1,2,3,4,8,12}	3, 3, 3	**yes, flat**
N=48	rho=0.6	3-7	{1,2,3,4,8,12}	4, 4, 4, 4, 4	**yes, flat**
N=96	rho=0.75	2-5	{12}	12, 12, 12, 12	**flat (grid of 1)**
N=96	rho=0.6	4-9	{12}	--, --, --, --, --, --	**not measurable**

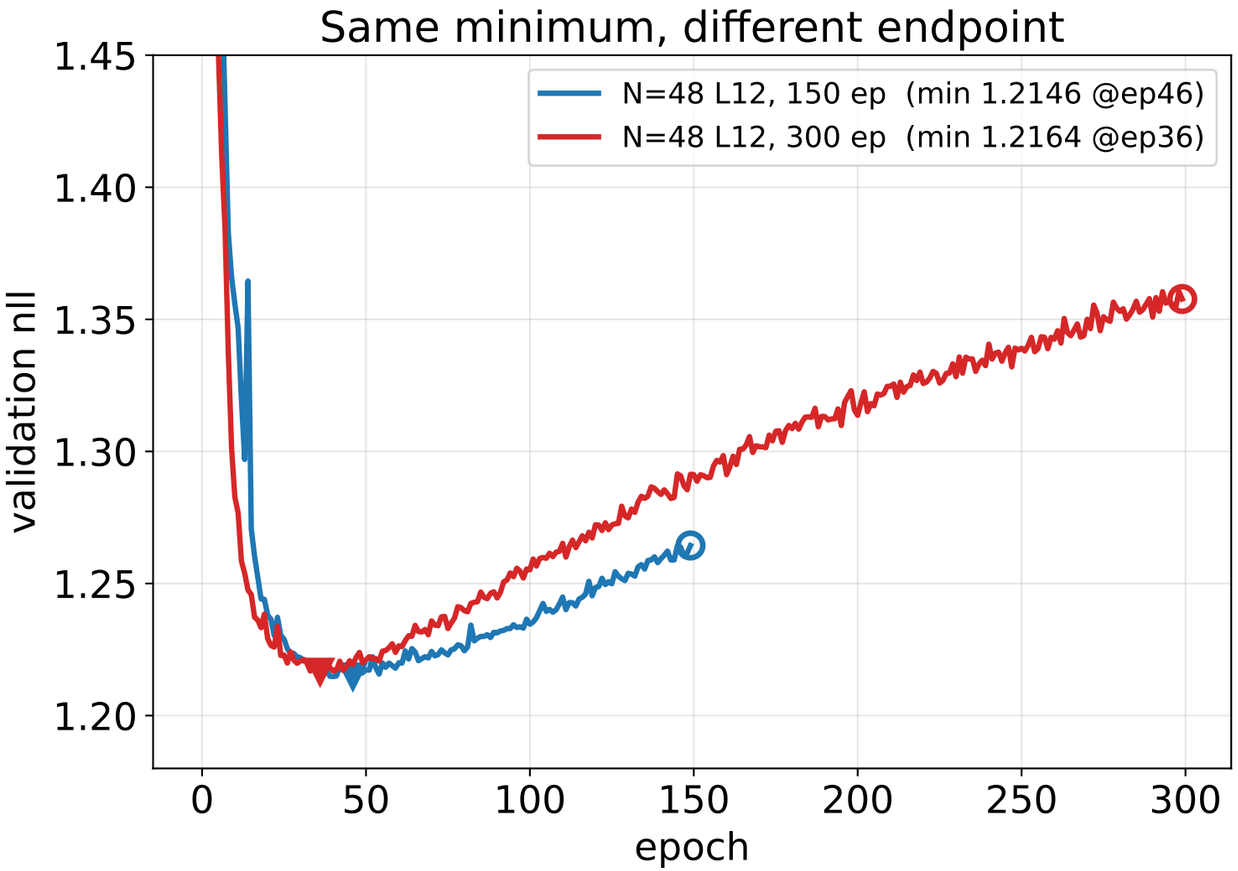
Read down the L*(T) column: within every row it is a constant. The N=96 rho=0.60 row is the honest hole -- no configuration measured there reaches 0.90 in any bin, so L* is undefined rather than large, and the dash is not a small number. The N=96 rows also searched a grid of one (L=12 only), so N=96's rho=0.75 entry is an upper bound, not a located knee: all it says is that twelve layers suffice. The cross-N comparison therefore rests on N=24 against N=48, where the full grid was searched at both sizes.

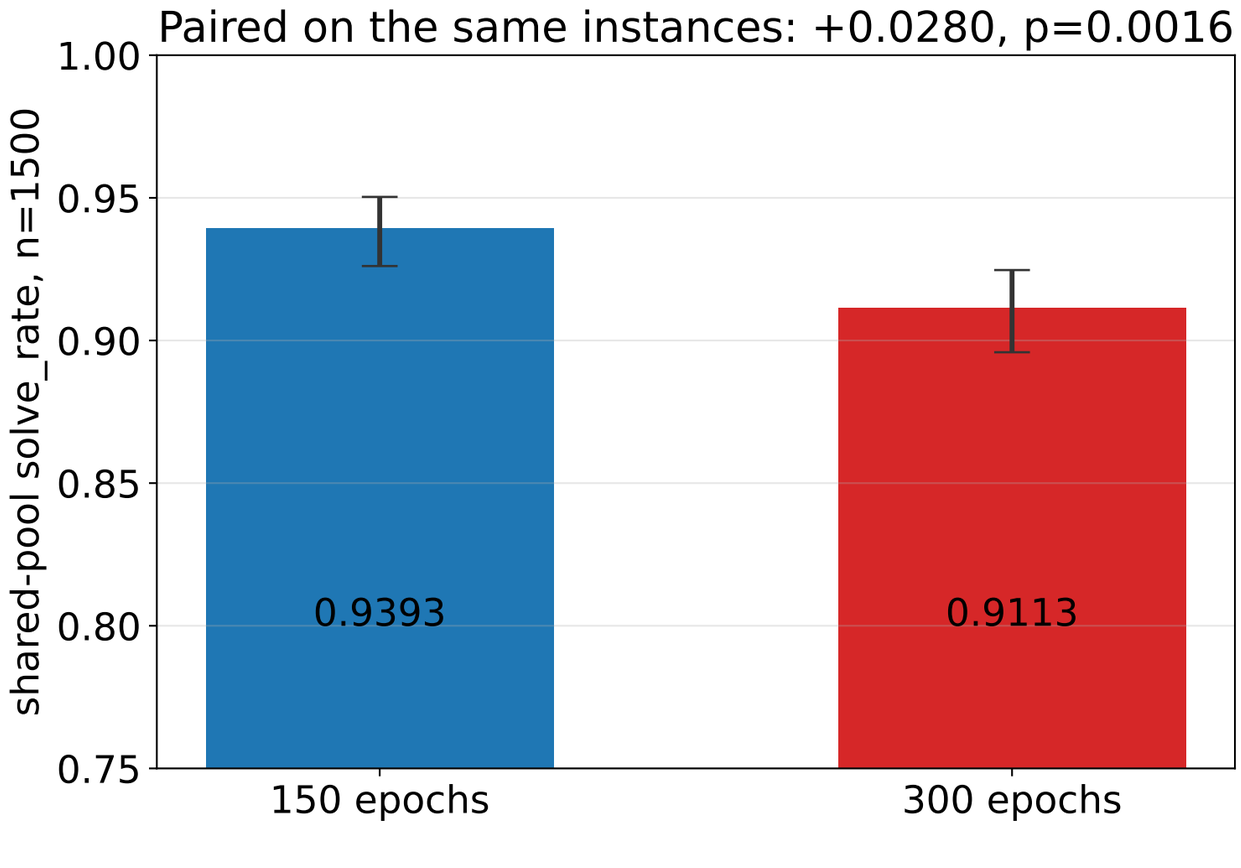


The curves that matter are the ones straddling the threshold. At N=24 that is L=3 (blue), which sits above 0.90 at every T from 2 to 5; at N=48 it is L=4 (green), above at every T from 3 to 7, while L=3 has dropped to 0.827-0.875 -- below the line at every depth, and by a nearly constant margin rather than a widening one. That is the shape of the whole result: when an extra layer does become necessary, it is necessary UNIFORMLY across T. A serial mechanism would instead show L=3 clearing the bar at shallow T and failing at deep T, crossing the line somewhere inside the plot. Nothing crosses. Caveat: each point is n=52-195 instances, so a single bin is resolved to roughly +-0.06 near 0.90 and no individual crossing would be trustworthy; what carries the claim is that a whole series sits on one side of the line, which is 4-5 bins agreeing at once.

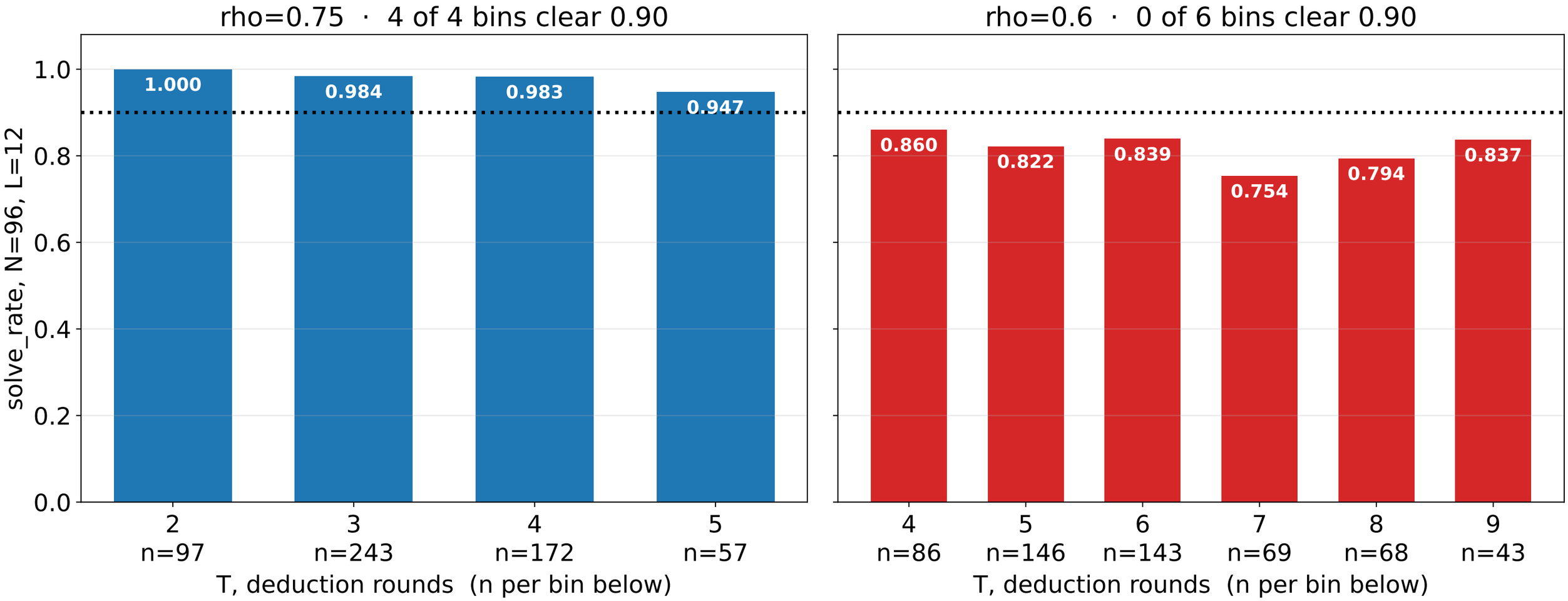


Three pre-registrations, all resolved. (a) L=3 scores below 0.90, i.e. L*(48)=4 and the knee rises with N: 0.8293 [0.8095, 0.8475], clear of the line with the whole interval to spare -- and it was a genuine 50/50 written down in advance. (b) the 150-epoch L=12 cell beats the 300-epoch one, paired p<0.05: +0.0280, 106-64 discordant, p=0.0016. (c) L=8 ties L=12: +0.0140, p=0.076, inside the 0.02 band and with L=8 the nominally better of the two, which is consistent with L=12 overfitting. Note what the ladder does past the knee -- it is flat from L=4 to L=12 to within 0.044, so the N=48 story is a knee, not a slope.

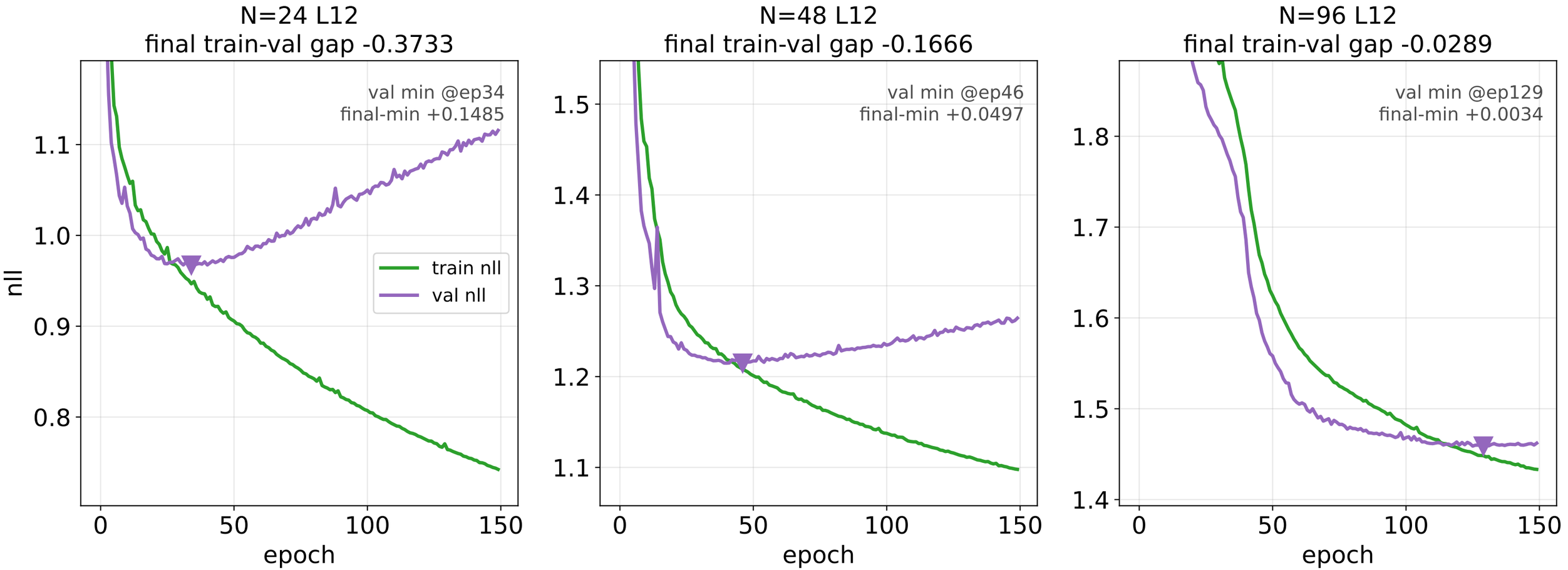




The layer scan's pre-registered gate FAILED: its L=12 cell was required to reproduce the 300-epoch baseline's 0.9033 within [0.865, 0.932] or the scan was to be reported and stopped as confounded with epoch budget. It came in outside -- but upward, 150 epochs beating 300, and on the shared pool the gap is +0.0280 with 106-64 discordant at p=0.0016, so it is real and not 300-instance noise. The left panel is the cause: both runs reach the same validation minimum (triangles, 1.2146@ep46 and 1.2164@ep36, so training is reproducible), then both overfit, and the 300-epoch run ends further from its minimum (circles, +0.1413 against +0.0497) because save_checkpoint keeps the final epoch, not the best one. That makes the confound cross-JOB, not within-scan -- every cell in the scan and the whole N=24 arm share the 150-epoch budget -- so it invalidates using the baseline as a reproducibility anchor, not the L* comparison. It is also the first time final-epoch checkpointing has been shown to cost measurable solve_rate rather than just val_nll.



The probe did its job by failing its own gate. It pre-registered that the recipe would clear 0.90 aggregate at N=96 so that L* would be measurable; it scores 0.8267 [0.8067, 0.8450] on the shared pool, and per bin it clears the threshold everywhere at rho=0.75 and NOWHERE at rho=0.60. Since L*(T) is defined by that threshold, a 36 h layer scan at N=96 would have returned a column of sub-threshold numbers at rho=0.60 with no L* in it -- and it would have run at half the data and two thirds the epochs, so the result would have been indistinguishable from its own budget. It stays unsubmitted. The other two pre-registrations held: the depth window widened again as intended, rho=0.60 now spanning T=4-9 against 3-7 at N=48 and 2-5 at N=24, which is why N=96 was worth probing at all.



Why the N=96 ceiling is probably NOT a data ceiling, and why that contradicts what the probe itself pre-registered as the next move. At N=24 and N=48 the saved model sits well past its validation minimum with train pulled below val -- the overfitting signature, where more data is the classical remedy and finding 6 says it has not saturated even at 80000. At N=96 the picture inverts: the minimum is at epoch 129 of 150, the last 30 epochs span 0.0043, and train ends 0.029 BELOW val, i.e. the model never fits its own training set. There is no generalization gap for more data to close. What there is instead is a constant learning rate: every run in this project used Adam at a fixed lr with no scheduler anywhere in the codebase. 49565433 tests cosine decay against 4x data on one pool, so the lever the probe pre-registered is still measured rather than dropped on an argument.

Against the previous best on the same eval, and what is not claimed

cell	solve_rate	Wilson 95%	n_train	epochs	source
N=24, L=4 (best on that ladder)	0.9760	[0.9670, 0.9826]	20000	150	scan n24
N=48, L=8 (best at this size)	0.9533	[0.9415, 0.9629]	20000	150	scan n48
N=48, L=3 NEW	0.8293	[0.8095, 0.8475]	20000	150	49523504
N=96, L=12 NEW	0.8267	[0.8067, 0.8450]	20000	150	49524030
project best: N=24, L=12, 4x data	0.9960	[0.9913, 0.9982]	80000	300	10-22 grid

Neither new cell is an improvement, and neither was meant to be: both were measured on purpose below the project best -- L=3 at N=48 to locate a knee from underneath, L=12 at N=96 to find out where a ceiling is. The 0.9960 row is a different problem size at 4x the data and 300 epochs, and is listed only so the two new numbers are not read as a regression.

```
commit 072f962 for both runs (identical to f8b15bc under spin_gnn/; 38d503a came after)
   seed=0 all training  ·  pool seed_start=5e10, n=1500, instance i at seed_start+i
         d_model=256, nhead=4, ff=1024, pre-norm, lr=1e-3 constant, batch=32
             degree=5, n_colors=4, rho=0.3, rho_max=0.95, removal hints
    49523504 ran 00:35:25, 49524030 ran 02:59:15, both ExitCode 0:0
```

One caveat on 'same eval': the pool is instance i at seed_start+i for the CONFIGURED problem size, so rows at equal N -- including the 0.9960 project best, same seed_start 5e10 and n=1500, from the 10-22 grid -- are literally the same 1500 instances and are paired-testable, while rows at different N share only the protocol. Every paired p-value in this deck is within one N. Open on purpose: N=96's L grid is one point, so its rho=0.75 entry bounds L* above by 12 rather than locating it, and the cross-N claim rests on N=24 against N=48 alone. L*(48)=4 at rho=0.60 but 3 at rho=0.75, so the level tracks density as well as size -- 'depth tracks graph size' is one data point at one density, not a law. What is solid is the negative: 15 bins across 4 fully searched series, none of them rising in T.